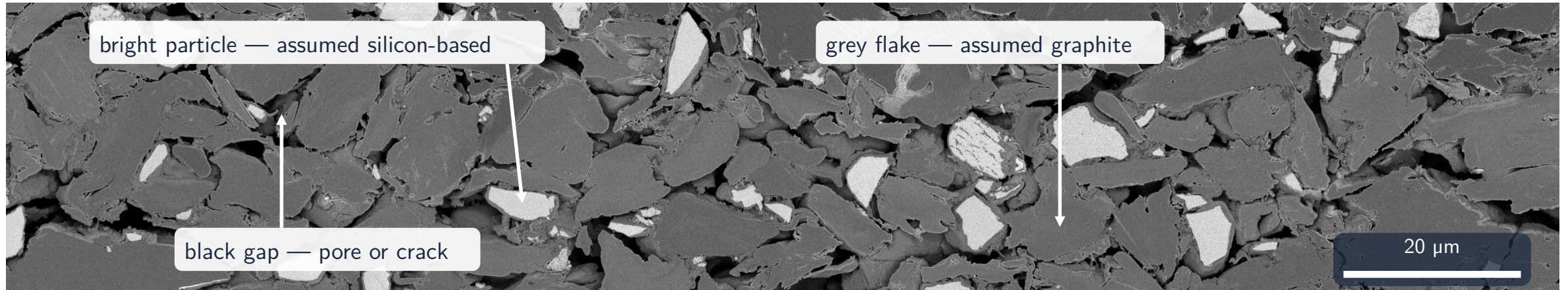


Do new anode batches match the approved one?

An explainable image-analysis pipeline for battery-electrode microscope photos



Batch 3 · img_cfe5vt7s · BSE detector · 25 nm per pixel · brightness stretched for display (same 1st–99.5th percentile stretch the pipeline uses)

THE PROBLEM

Does a new delivery of anode coating match the approved Batch 3?

THE DATA

31 cross-section photos
(7 + 7 + 17 locations),
3 detectors each

WHAT IT PRODUCES

Pixel maps, 20 repeatable
measurements, shifts vs
Batch 3 with uncertainty

WHO COULD USE IT

Quality and process
engineers, and suppliers
deciding what to re-check

What it can do today

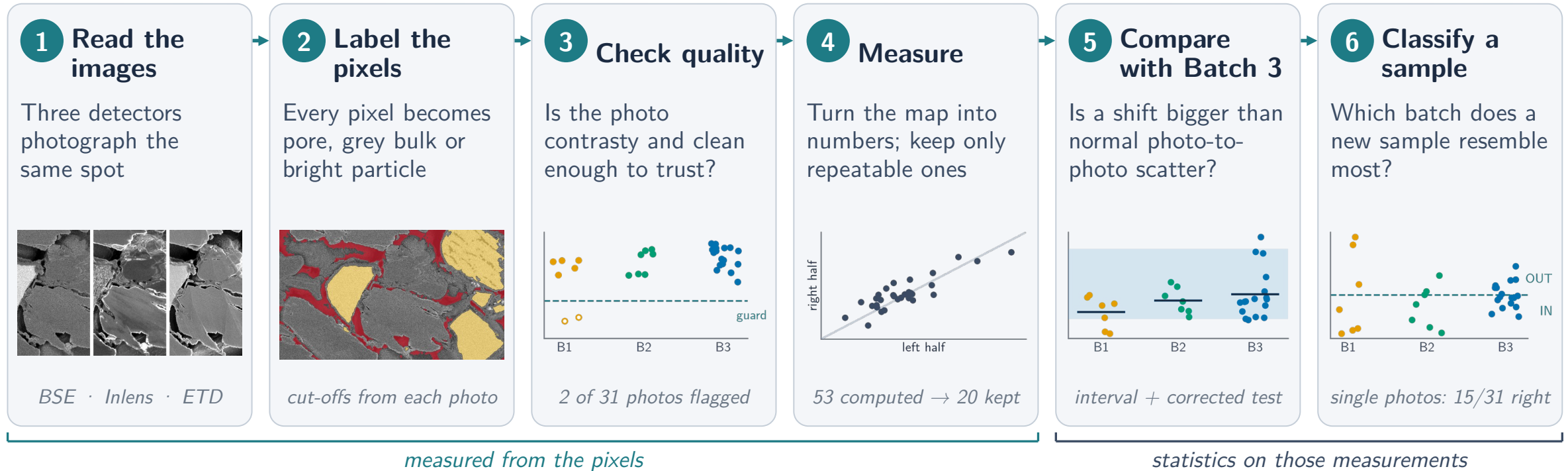
● **Image comparison: works**
measured shifts come with intervals

● **Anomaly / batch call: not yet**
single photos: 15 of 31 correct

● **Battery simulation: illustrative**
optional, model-dependent branch

How the algorithm works

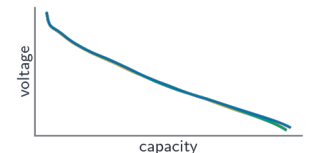
One fixed recipe, applied identically to every photo — six steps from picture to comparison



Assumption from step 2 on: bright = silicon-based, grey = graphite. This is read from brightness, not confirmed by chemical analysis (e.g. EDS).

Optional branch — a battery model

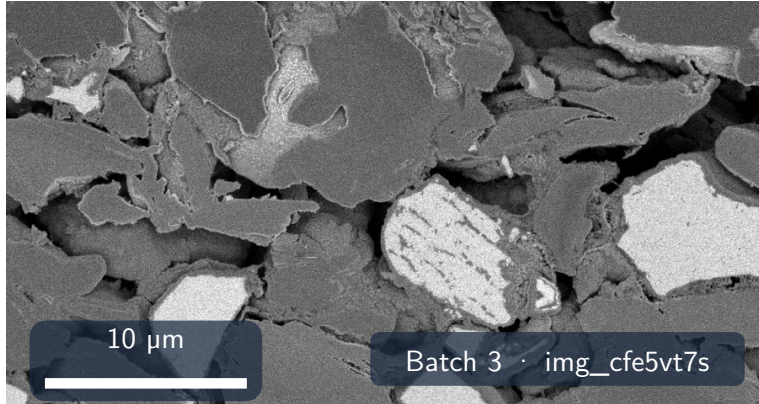
Image porosity, phase shares and particle sizes feed a published cell model (PyBaMM DFN); everything else is assumed.



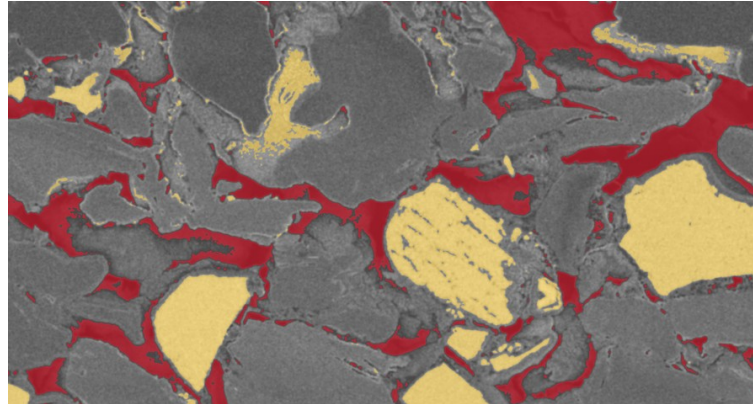
How we segment the images

One recipe for every photo; the cut-offs come from each photo's own brightness histogram

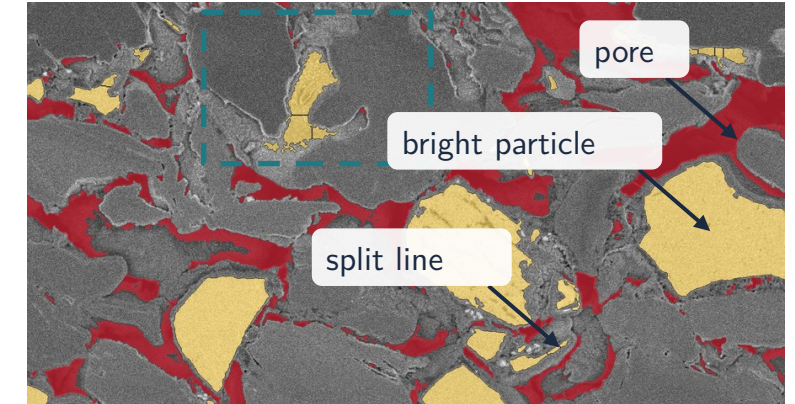
1 · Original photo (BSE)



2 · Brightness cut-offs, before clean-up



3 · Final labels



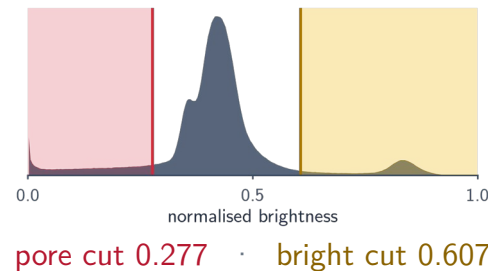
■ pore / crack (dark) ■ bright particle — assumed silicon-based ■ grey bulk — assumed graphite ■ particle outline (watershed)

What the recipe does

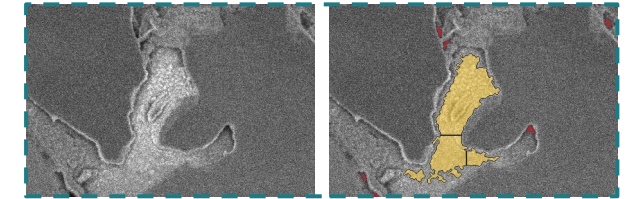
- 1 **Stretch & smooth:** 1st–99.5th percentile, 1.5-px blur
- 2 **Two cut-offs:** darkest group → pore, brightest → particle
- 3 **Clean up:** drop specks, fill holes, add Inlens cracks
- 4 **Split touching particles:** watershed on distance to edge

Why: transparent, no training, same settings for all photos

This photo's histogram



Close-up: an ambiguous object



Textured light-grey object: only partly labelled bright, and split into 4 'particles'.

Main limitation: brightness is not chemistry — 'bright = silicon' is unconfirmed, low-contrast photos inflate the bright class, and gaps under ~50 nm (2 px) are invisible.

The markers that matter — and what we can conclude

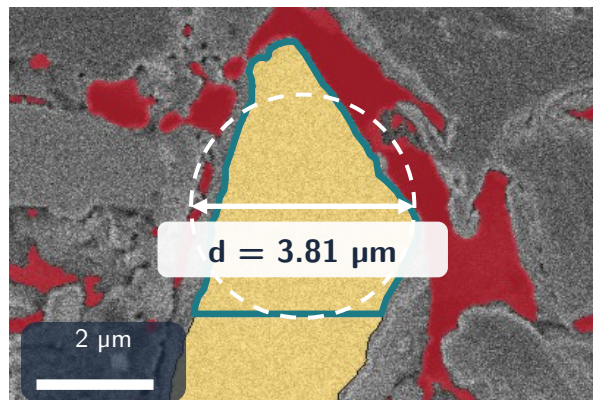
Five of the 20 retained measurements, chosen for reliability and physical relevance

Marker (plain English)	What the image measurement tells us	Why it might matter for a battery	What we can actually conclude
Visible pore space pore_frac	Share of the section that is dark: gaps wider than ~50 nm	Pores carry the electrolyte: more pore space speeds ion transport but stores less energy per volume [1]	Tentative. Batch 1 is −14% vs Batch 3 (interval excludes zero, not significant after correction). Batch 2: no detected difference.
Bright-particle share silicon_frac	Share of bright particles — assumed silicon-based, not chemically confirmed	Silicon holds ~10× more lithium per gram than graphite but swells up to ~280% [2]	No difference detected once two low-contrast Batch 1 photos are set aside (−2%); with them Batch 1 reads +45%.
Bright-particle size si_d50_um	Median diameter of the bright particles in a photo (equal-area circle)	Smaller particles crack less when they swell [3] but expose more surface to side reactions [4]	Undetermined: Batch 3 averages 0.76 μm; Batch 1 −15%, Batch 2 −12%, both within the noise. A 2-D cut understates true size.
Pore space next to particles ring_porosity_250nm	Pore share of a 250-nm shell around each bright particle	Nearby voids can absorb swelling and let electrolyte in [5] ; too much may weaken contact	No difference detected: 8–10% on average in every batch.
Graphite flake alignment gr_chord_ratio_hv	Horizontal vs vertical run length through the grey flakes	Flat-lying flakes lengthen the ion path through the coating thickness [6,7]	Shared feature: about 1.2 in every batch (flakes ~20% wider than tall); equivalent within ±10%.

Comparisons: mean shift vs Batch 3 with a bootstrap 95% interval, Mann–Whitney test, Holm correction over 20 markers; 7 vs 17 photos. Variable names link to their definitions in main.
[\[1\] Singh et al. 2016](#) [\[2\] Obrovac & Chevrier 2014](#) [\[3\] Liu, X. H. et al. 2012](#) [\[4\] Wu & Cui 2012](#) [\[5\] Liu, N. et al. 2012](#) [\[6\] Ebner et al. 2014](#) [\[7\] Billaud et al. 2016](#)

What these markers look like

One typical Batch 3 photo (img_cfe5vt7s) · all outlines are algorithm labels, none expert-confirmed



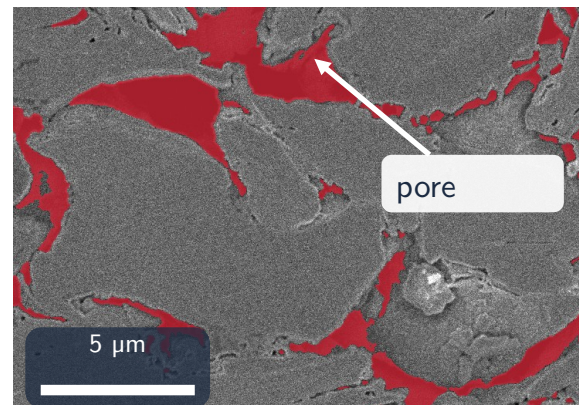
Particle size

MAIN METRIC

d = 3.81 μm

diameter of a circle with the same area as this particle's cut

Photo median over 327 particles: 0.66 μm — many small ones pull it down.



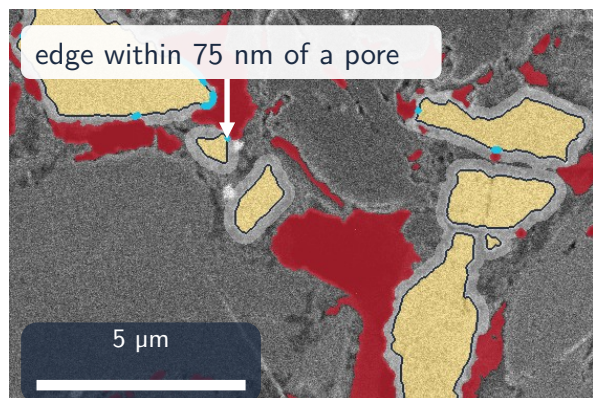
Visible pore space

MAIN METRIC

10.8%

of this photo's area is pore

Red = darker than the pore cut-off: gaps wider than ~50 nm. Finer pores are invisible.



Particle–pore contact

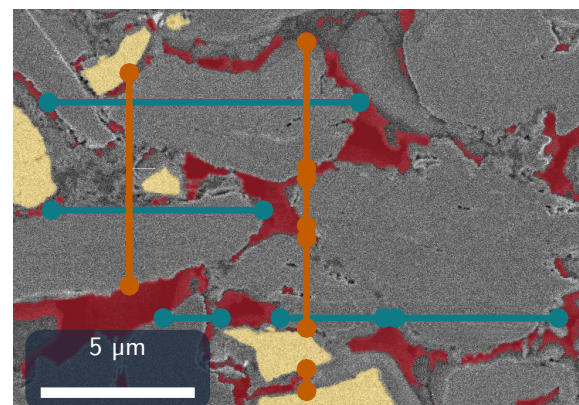
2 MAIN METRICS + 2 DERIVED

23% of particles touch a pore
(derived)

49% of bright area is in them
(derived)

1.7% of particle edge touches a pore
(main)

8.6% of a 250-nm shell is pore
(main)



Flake alignment

MAIN METRIC

ratio 1.21

mean run **across** 3.75 μm vs **down** 3.09 μm

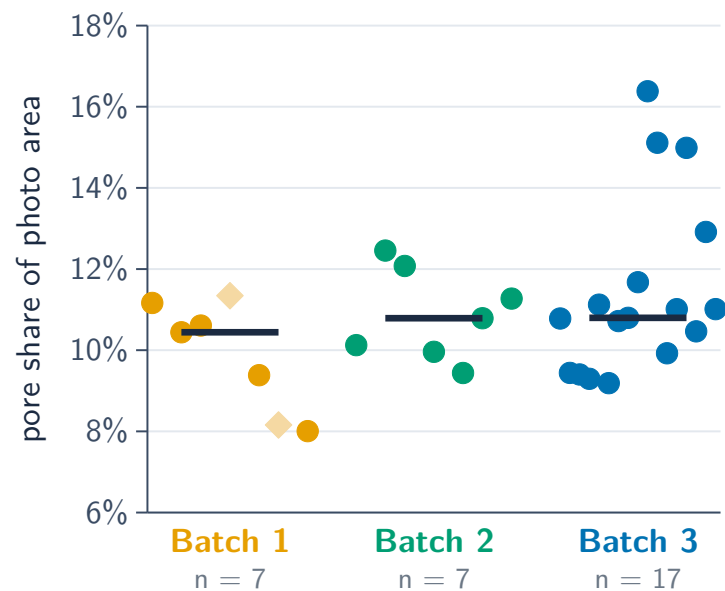
Runs through the grey flakes are longer sideways: the flakes lie flat.

‘Contact’ here is a 2-D distance in one section (edge within 75 nm of a pore) — not wetting, electrical contact or electrochemical activity. Main = a metric computed by main; derived = computed for this slide from main's per-particle table.

The key measurements, photo by photo

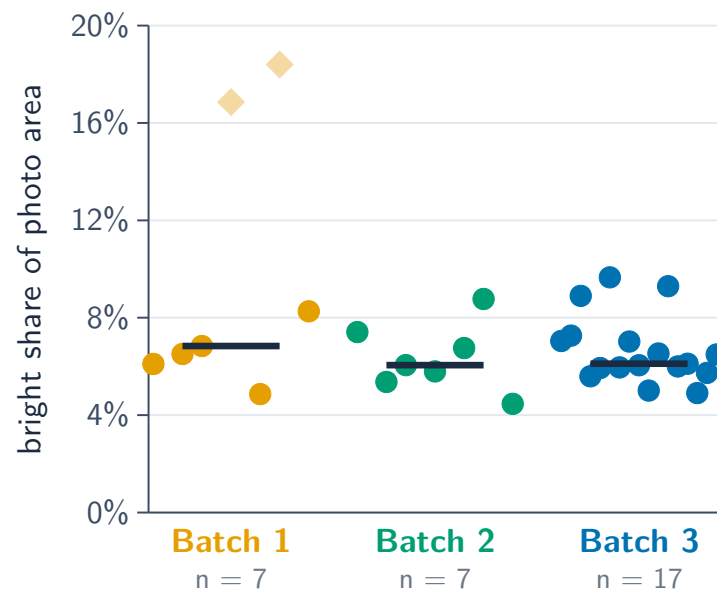
One dot per photo (one imaged location) · lines = batch medians · light diamonds = flagged by the quality guard

A · Visible pore space



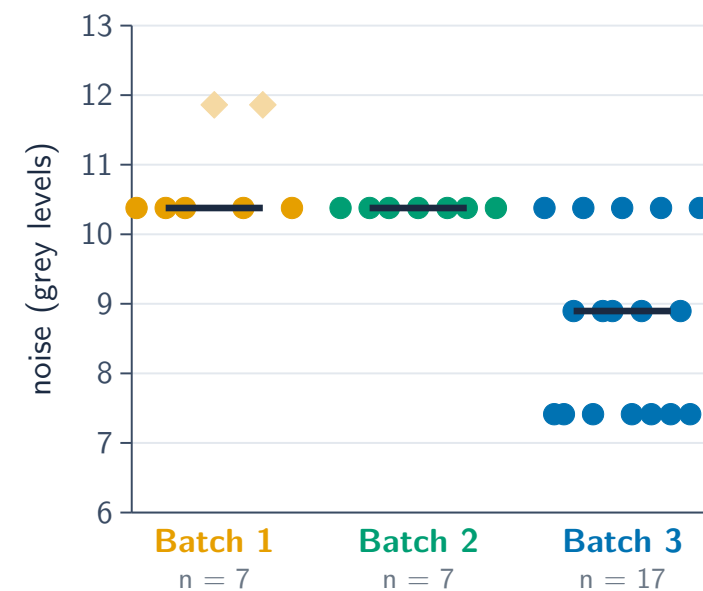
Batch 1 sits lower on average (—14% vs Batch 3), but its photos overlap Batch 3's: a tentative shift.

B · Bright-particle share



Two low-contrast Batch 1 photos read 16.9% and 18.4%; the other 29 lie between 4.5–9.7%. Without them, no batch differs.

C · Image noise level



Batch 1 and 2 photos are noisier than most Batch 3 photos. Texture, the one marker passing the corrected test, follows this noise ($r = 0.91$).

Imaging is entangled with batch: 12 of the 17 Batch 3 photos were taken at lower noise than any Batch 1 or 2 photo, so batch and microscope-settings differences cannot be fully separated.

Results by batch — and the next decision

Batch	Main measured observations	Status	Follow-up justified by the evidence
Batch 1 7 photos	▪ Less visible pore space (−14%) and fewer large gaps (−71%) ▪ Bright share +45% due to two low-contrast photos (−2% without them) ▪ Texture +18%, which tracks image noise	Tentative no material shift survives the correction	Re-image the two flagged locations; add 5+ locations with Batch 3's microscope settings
Batch 2 7 photos	▪ No established structural shift: finest pores −7% and flake texture order −8% are tentative ▪ Texture +15%, which tracks image noise	No difference established	No action on structure; keep imaging settings matched to Batch 3
Batch 3 17 photos reference	▪ Defines the normal range for every marker ▪ Held out, 7 of 17 of its own photos fall outside its single-photo envelope	Reference envelope too tight for single photos	Judge groups of photos; widen the reference with more known-good batches
New batch results being generated	<i>Pending — the run on the new delivery is in progress; its results are inserted here.</i>	Pending	Write down the expected answer before the results are revealed

Take-home: the pipeline measures structure reproducibly and catches unreliable photos. The batches look broadly alike — Batch 1's lower pore space is tentative — and single-photo batch calls are not yet reliable.

Most useful next validation: a blind test — photograph 5 or more locations of the new delivery with Batch 3's microscope settings, write down the expected answer first, have an expert fill main's 300-point check sheet, and confirm the bright particles chemically (EDS).

Simulation branch (model scenarios, not measurements): at 2C with measured porosity, Batch 1/2 run −2.7%/−2.9% vs Batch 3; with porosity set to 30% they match; at 1C and slower every run stops at the simulation time limit (5.25 Ah).